

4-3-1: Examining IPv4 Addressing

At the end of this episode, I will be able to:

1. Identify and explain the process and components of IPv4 addressing, given a scenario.

Learner Objective: *Identify and explain the process and components of IPv4 addressing, given a scenario.*

Description: In this episode, the learner will examine IPv4 addressing in TCP/IP networks. We will explore the process and components that make up an IPv4 address.

- What do we mean by IPv4 addresses?
 - o Unique identifiers for devices on a network.
 - o 32 bits divided into four octets, like 192.168.1.1.
 - o Each of the four octets in the address can have a value from 0 to 255, providing a range of possible addresses.
 - Why is understanding binary important for IP addressing and subnetting?
 - o Binary is crucial because IP addresses, though often represented in decimal, are fundamentally processed in binary by network devices.
 - o This knowledge is key to understanding network addressing and how subnetting works at a technical level.
 - What role does a subnet mask play?
 - o A subnet mask works alongside an IP address to define which part of the address identified or belongs to the network and which part refers to the host.
 - o Determine the subnet to which an IP address belongs.
 - How do we tell what network we are on?
 - o The network ID (or network portion) of an IP address, determined by the subnet mask, indicates the specific network a device is on.
 - o The host ID (or node ID), which is the part of the IP address not covered by the subnet mask, identifies the specific device within that network.
 - o Understanding the network and host portions of an IP address is crucial for determining both the network a device belongs to and its unique identifier within that network.
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- Additional Resources:
 - o If applicable